

Avanthika Vuppala

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Interests: Aerospace Systems Design, Structures, Propulsion, Advanced Materials, Robotics

EDUCATION

Rutgers University

New Brunswick, NJ

Bachelor of Science in Aerospace Engineering, Minor in Computer Science

Sept 2023 – May 2027

Relevant Coursework: Statics, Dynamics, Mechanics of Materials, Thermodynamics, Fluid Mechanics, Differential Equations, Mechatronics, CAD in Mechanical Engineering, Dynamic Systems and Controls, Aerospace Structures, Aerospace Materials

EXPERIENCE

Undergraduate Research Assistant

Feb 2025 – Present

Hybrid Micro/Nanomanufacturing Laboratory - Rutgers University

New Brunswick NJ

- Improve energetic material structural reliability by synthesizing and characterizing 10+ MINET composite formulations with controlled pore architectures for a DARPA-funded propulsion program
- Quantify mechanical performance trends by conducting 20+ tests and analyzing 200+ data points to evaluate stiffness, strength retention, and fracture behavior
- Support next-generation solid propellant development by generating validated experimental data and maintaining structured technical documentation informing material selection and combustion response

Research Internship

Jun 2025 – Aug 2025

Advanced Controls Lab - Rutgers University

New Brunswick, NJ

- Developed ROS 2 applications in Python and C++ to program an iRobot Create 3 mobile robot for autonomous navigation and motion control experiments
- Integrated sensors and implemented localization and path-planning algorithms to support robotics research in autonomous systems and control
- Performed system testing and troubleshooting while documenting software architecture, experimental procedures, and results in a technical report for future research use

Science Team Lead

Jan 2025 – Apr 2025

NASA L'SPACE Proposal Writing and Evaluation Experience Academy

Remote

- Defined a feasible microgravity refueling science concept by leading development of the On-Site Refueling Device using magnetic docking with a 7-member multidisciplinary team
- Translated mission science into engineering requirements by authoring a 30-page proposal including success criteria, TRL assessments, and a Science-Technology Traceability Matrix
- Validated multidisciplinary system feasibility by performing heritage system reviews and technology trade studies with multiple engineering subteams

PROJECTS

On-Site Refueling Device - Universal Magnetic Docking Interface

Jan 2025 – Apr 2025

- Designed a spacecraft refueling concept using a self-aligning magnetic docking interface to simplify orbital propellant transfer
- Developed system architecture and engineering requirements with a multidisciplinary NASA L'SPACE team
- Evaluated design tradeoffs to improve docking reliability and reduce mechanical complexity

Portable Refracting Telescope and Tripod System Design

Dec 2025

- Designed a 65 lbf load-rated structural support system validated using analytical calculations and SolidWorks FEA
- Developed multi-level CAD assemblies and fabrication drawings supporting mechanical integration
- Evaluated load cases, materials, and manufacturability to improve structural reliability

TECHNICAL SKILLS

Engineering: SolidWorks, Siemens NX, AutoCAD, MATLAB, Arduino, Python, C++

Analysis: CAD Modeling, FEA, Mechanical Testing, Data Analysis, Engineering System Modeling

Tools: Microsoft Office (Excel, Word, PowerPoint), Google Workspace, Outlook